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Stocks price prediction based on optimized echo state network by sparrow search algorithm

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Abstract
The stock market reflects the level of economic development of a country to a certain extent. Therefore, stock price forecasting has become a popular research issue. More and more attention has been paid to the establishment of stock price forecasting models. However, the stock market is a nonlinear system filled with high noise, and the traditional fundamental analysis, technical analysis and ARIMA model cannot achieve the ideal prediction result. Although traditional neural networks have good adaptability and nonlinear approximation capability, but they also have drawbacks, such as slow training speed and easiness to fall into local optimization. The echo state network–sparrow search algorithm (SSA–ESN) model employed in this paper uses the sparrow search algorithm to optimize the parameters of echo state network (ESN). It has not only better prediction result but also inheriting the excellent properties of ESN. The SSA–ESN model is used to predict several stocks price, and the results are compared with those of BP neural network, Elman neural network, LSTM and echo state network. And the prediction accuracy of SSA–ESN model is significantly better than those of the other four prediction models.

Keywords Stocks price · Sparrow search algorithm · Echo state network

1 Introduction

The stocks market serve as a barometer of a country's economic development, to some extent. In recent years, numerous scholars have conducted diverse studies on the stocks market, proposing a range of analytical and forecasting methods to address its unique characteristics. These methods include fundamental analysis, technical analysis and time series analysis. However, the stocks market is a dynamic, nonlinear system with high noise, which poses challenges for the application of traditional prediction methods.

Neural networks have gained popularity in stocks price prediction due to their exceptional adaptability and nonlinear approximation capabilities. While traditional neural networks like BP neural networks exhibit impressive performance, they also display significant limitations such as slow training speeds, tendency toward local optimization and vulnerability to noise.

Computational intelligence is a collective term for algorithms developed by scholars through computer implementations of biological intelligence. This field provides numerous optimization methods for time series prediction and has numerous successful applications in dealing with massive data simulations. However, individual intelligent algorithms often fall short in their applicability to a wide range of problems, exhibiting limitations in both theory and practice. Therefore, combining intelligent optimization algorithms with neural networks has become the mainstream approach for stocks price prediction among scholars today. This paper adheres to such an idea, based on a special neural network–echo state network, using the sparrow search algorithm to optimize its learning rate and regularization coefficient, to improve the prediction accuracy of the echo state network on the basis of trying to maintain its original operational efficiency. SSA–ESN shows a greater advantage in prediction accuracy compared to BP neural networks, Elman neural networks, LSTM and echo state networks.

In recent years, different methods have been proposed for stocks price forecasting. The ARIMA and GARCH models had been widely used for financial market forecasting [1]. However, due to the complexity and nonlinearity of the time series of stocks price, many traditional models cannot

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predict stocks price perfectly. So new models and strategies are emerging. The good nonlinear approximation ability of neural networks can better solve the high noise and nonlinear problems existing in the stock market. Adebiyi et al. conducted a comparative test on stock data using ARIMA model and artificial neural networks [2]. Lu et al. introduced computational intelligence combined with BP Neural Network to improve the prediction performance of BP Neural Network [3]. Xiao et al. proposed an integrated model that combines cumulative autoregressive moving average and least squares support vector machine (ARIMA-LS-SVM) to predict stocks price trends [4]. Rezaeia et al. combined LSTM, CNN and CEEMD or EMD algorithms to improve the accuracy of stocks price prediction [5]. Emam et al. propose a new machine learning technique that uses a confidence rules-based expert system (BRBES) for technical analysis and combines the concept of Bollinger bands to predict stock prices for the next 5 days [6]. Shetty and Ismail combined nonsmooth trends, lagged variables and ERNN to capture linear and nonlinear structures and improve stock price forecasting accuracy [7]. Mehdi Khashei and Zahra Hajirahimi staged stocks price prediction by proposing two hybrid models for stocks price prediction: the ARIMA-MLP model and the MLP-ARIMA model [8]. Bose et al. emphasized the importance of dimensionality reduction and combined multivariate adaptive regression splines (MARS) and deep neural networks (DNN) to predict the closing price of stocks [9]. Wang et al. proposed an interval variational mode decomposition (FIVMD) method with feedback mechanism to model interval trends and residuals, respectively, through optimization algorithms and deep learning models to improve prediction accuracy [10]. Du used arbitrage pricing model to establish the relationship between various system factors and adjusted stock prices [11]. Kumar et al. proposed a hybrid long short-term memory network model based on adaptive particle swarm optimization to predict three major stock indexes in the short and long term [12]. Teng et al. proposed a multi-scale local cue and attention-based LSTM model (MLCA–LSTM) to capture potential price trend patterns [13]. Wang et al. introduced time series technology to conduct comprehensive modeling of stock price data and established TSC model to detect the uncertainty of data changes [14]. Liu et al. proposed a new stock price prediction model VML, which uses MAML algorithm and long short-term memory (LSTM) network to predict subsequences [15].

The literature review indicates that it has become a mainstream to combine multiple models or to employ computational intelligence to optimize the hyperparameters of neural networks to improve the accuracy of stocks price prediction. However, the most convenient network is prioritized when all networks are equally feasible. Echo state network is a new recursive neural network algorithm proposed by Jaeger in 2001. Jaeger used the ESN to forecast the time series and got better results. Jaeger compared it with forward type neural network and traditional recursive neural network. In the meanwhile, Jaeger gave the algorithmic rules and training mode of ESN [16]. It is a special kind of neural network with good short-term memory, and its simple training process and unique way of constructing the reservoir pool make its training efficiency greatly improved, which is widely used in time series prediction. ESN has a good function approximation capability, which can effectively solve the serious nonlinearities of the stocks market. ESN has a great advantage over traditional neural networks in terms of stability, global optimality, local minimum problem and the complexity of the training process. Compared with traditional neural networks, ESN has great improvement.

Skowronski and Harri applied ESNs to automatic language recognition classification [17]. Lin argued that the random selection of the initial values of network weights made the network prediction results unstable and used the Hurst index to set the weights of ESN [18]. However, Ishu et al. argued that ESN does not need to rely on a large amount of experience as traditional neural networks do [19]. Zhang proposed a generalized model for short-term stocks price prediction based on echo state network [20]. Zhang et al. used a synthetic model of phase space reconstruction, ESN, and moving average convergence/divergence (MACD) for stocks prediction and compared it with other models, such as backpropagation neural network (BPNN), support vector machine (SVM), hidden Markov model (HMM) and programmed neural network (PNN), to validate the high efficiency of the composite model consisting of ESN for stocks price prediction [21].

Sparrow search algorithm (SSA) is a novel intelligent optimization algorithm proposed by Xue et al. based on the predatory behavior of sparrow population [22]. Compared with traditional intelligent algorithms, it has a strong global search capability. Liu et al. introduced the sparrow search algorithm to optimize the hyperparameters of LSTM model to improve the accuracy of stocks trend prediction [23]. Sparrow search algorithm to optimize neural network for regression prediction is also applied in other fields. Liu et al. combined SVM model and SSA to predict the calorific value and nitrogen content based on feedstock properties and pyrolysis conditions [24]. Liao proposed a load forecasting model incorporating improved sparrow search algorithm and long and short-term memory neural network (LSTM) [25]. Tan et al. optimized PSR and ESN by combining TVFEMD, IASSA and achieved more accurate wind power prediction [26]. Zheng et al. proposed BP neural network model optimized based on SSA. Compared with the traditional BP neural network, it has higher accuracy and better fitting result in ship trajectory prediction [27].

To summarize, although traditional neural networks such as BP neural networks reflect advantages in predicting stocks

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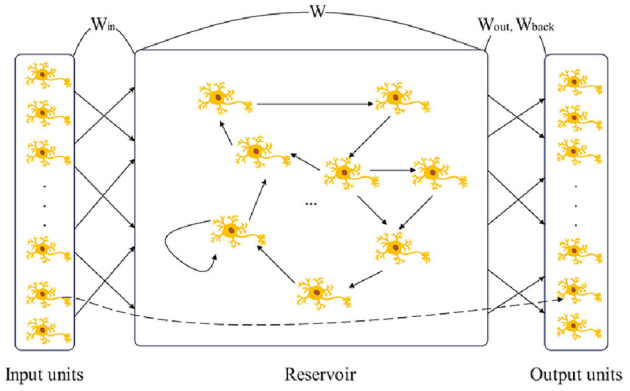


Fig. 1 Structure of echo state network (ESN) model

price compared to traditional prediction methods, they are very sensitive to noise and are easy to fall into local optimal points. A great deal of previous studies by scholars have proved that ESN has obvious advantages in the stock price prediction problem, such as simple model construction and high prediction accuracy. The SSA–ESN model used in this paper solves the shortcomings of traditional neural networks such as BP in stocks price prediction to a certain extent. And it improves the prediction accuracy of ESN on the basis of maintaining the computing efficiency of echo state network.

The rest of the paper is organized as follows. In Sect. 2, the basic concepts of ESN and SSA are briefly introduced, as well as the modeling steps for optimizing ESNs using SSA. In Sect. 3, the basic assumptions of the paper, data preprocessing and the experimental configuration are presented. In Sect. 4, the prediction results of using SSA to optimize ESN to predict ten stocks in China and the USA are shown, and the predictive effectiveness of the model is evaluated in terms of both the root mean square error (RMSE) and the mean absolute percentage error (MAPE). In Sect. 5, the prediction performance of the BP neural network, Elman neural network, LSTM, ESN and SSA–ESN are compared using the same dataset as in Sect. 4. In addition, the predictive performance of the SSA–ESN model is tested for robustness under different time frames and market conditions. Section 6 contains the concluding remarks.

2 SSA–ESN model construction

2.1 Echo state network

As shown in Fig. 1, the basic structure of ESN is an input layer, a reservoir pool and an output layer.

The core structure of ESN is a randomly generated reservoir pool, similar to the hidden layer of a general neural network. But its size is much larger than that of a traditional neural network hidden layer, usually around tens or hundreds. Its internal neurons are sparsely connected, and the sparsity SD is generally taken as 2%-5%.

At moment n, the sequence of input cells is $u(n) = [u_1(n), u_2(n), \ldots u_k(n)]$, the sequence of states of the reservoir pool is $x(n) = [x_1(n), x_2(n), \cdots x_N(n)]$ and the sequence of output cells is $y(n) = [y_1(n), y_2(n), \cdots y_L(n)]$.

The update equation for the state of the reservoir pool is

$$x(n+1) = f(W_{in}u(n+1) + Wx(n) + W_{\text{back}}y(n)), \tag{1}$$

where W_{back} is not required. In this paper, we let $W_{\text{back}} = 0$ to obtain the update equation for the state of the reservoir pool as

$$x(n+1) = f(W_{\text{in}}u(n+1) + Wx(n)), \tag{2}$$

where W_{in} refers to the feedback matrix from the input layer to the reservoir pool and W is the feedback matrix within the reservoir pool.

The output of ESN is given by the following equation,

$$y(n+1) = F(W_{\text{out}}(u(n+1), x(n), y(n))). \tag{3}$$

ESN operates efficiently since it only needs to learn and train W_{out}. W_{in} and W remain constant once they are determined. However, there is no strict determination criterion for the learning rate and regularization coefficients, and the random selection of the learning rate and regularization coefficients makes ESN output unstable. The SSA–ESN model constructed in this paper solves this problem effectively.

ESN has the function of transiently memorizing the input signal $u(n)$, i.e., the echo state. The condition for ESN to have an echo state is that the spectral radius of the internal weight matrix W of the reservoir pool is less than 1. That is, it is required that the maximum spectral radius of eigenvalues of W is satisfied

$$|\lambda|_{\max} < 1. \tag{4}$$

For the randomly generated internal weight matrix of the reservoir pool, find its maximum value of the updation of weight matrix is

$$W_{\text{new}} = \frac{1}{|\lambda|_{\max}} W, \tag{5}$$

where W_{new} is the new internal weight matrix.

2.2 Sparrow search algorithm (SSA)

SSA algorithm is a swarm intelligence optimization algorithm based on sparrow foraging and predator avoidance behaviors. Sparrows are divided into discoverers and followers when foraging. The discoverers are responsible for

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finding food in the group, providing foraging areas and directions to the whole sparrow group. And the followers follow the discoverers to obtain food. In order to obtain food, sparrows can generally adopt two behavioral strategies for foraging: discoverer and follower. Individuals within the population will monitor the behavior of other individuals within the population, and aggressors within the population will increase their predation rate by competing for food resources with their high-intake peers. In addition, sparrow populations will engage in antipredatory behavior when they recognize danger.

In SSA, each individual (sparrow) has a position that represents its location in the search space. In each iteration, the position of each individual is updated based on the fitness value. Because the discoverer is responsible for finding food for the entire sparrow population and providing foraging directions to all followers, the discoverer can obtain a larger foraging search range than the followers.

During each iteration, the location update of the discoverer is described as

$$X_{i,j}^{t+1} = \begin{cases} X_{i,j}^{t} exp\left(-\frac{i}{\alpha \text{iter}_{\max}}\right), & R_2 < \text{ST}, \\ X_{i,j}^{t} + QL, & R_2 \geq \text{ST}. \end{cases} \tag{6}$$

In (6), t is the current iteration number; $\text{iter}_{\max}$ is the maximum iteration number; $X_{i,j}^{t}$ is the position information of the ith sparrow in the jth dimension; $\alpha \in 0, 1$is a random number; R_2 and ST denote the warning value and the safety value, respectively, where $R_2 \in [0, 1]$, $\text{ST} \in [0.5, 1]$; Q is a random number obeying the normal distribution; L is the matrix of $1 \times d$, where each element of the matrix is all 1. For $R_2 < \text{ST}$, it indicates that there are no predators around the foraging environment at this time, and the discoverer can perform extensive search operations; for $R_2 \geq \text{ST}$, it means that some sparrows in the population spot a predator and alert others in the population, and then all need to quickly fly to other safe places to feed.

The follower's location updation is described as

$$X_{i,j}^{t+1} = \begin{cases} Q \exp\left(\frac{X_{\text{worst}}^{t} - X_{i,j}^{t}}{i^2}\right), & i > n/2, \\ X_{p}^{t+1} + \left|X_{i,j}^{t} - X_{p}^{t+1}\right| A^{+} \cdot L, & \text{else}, \end{cases} \tag{7}$$

In (7), X_{P} is the optimal position occupied by the current discoverer; X_{worst} denotes the current global worst position; A is a $1 \times d$ matrix, where each element is randomly assigned at a value of 1 or -1, and $A^{+} = A^{\text{T}}\left(AA^{\text{T}}\right)^{-1}$, where A^{+} is the pseudo-inverse matrix. When $i > n/2$, it indicates that the ith follower with a lower fitness value does not get food and is in a very hungry state, and then needs to fly to other places to forage for food to get more energy.

When aware of danger, sparrow populations engage in antipredatory behavior, the mathematical expression for which is

$$X_{i,j}^{t+1} = \begin{cases} X_{\text{best}}^{t} + \beta \left|X_{i,j}^{t} - X_{\text{best}}^{t}\right|, & f_i > f_{\text{g}} \\ X_{i,j}^{t} + K \left[\frac{\left|X_{i,j}^{t} - X_{\text{worst}}^{t}\right|}{(f_i - f_{\text{w}}) + \varepsilon}\right], & f_i = f_{\text{g}} \end{cases} \tag{8}$$

In (8), X_{best} is the current global optimal position; β is a step control parameter, which is a normally distributed random number with mean 0 and variance 1; K is a random number indicating the direction of the sparrow's movement as well as the step control parameter, and $K \in [-1, 1]$; f_i is the current fitness value of the individual sparrow; f_{g} and f_{w} are the current global optimal and worst fitness values, respectively; and ε is the smallest constant in order to avoid the denominator to be zero. For simplicity, for $f_i < f_{\text{g}}$, it means that the sparrows are at the edge of the population and are extremely vulnerable to predators; for $f_i = f_{\text{g}}$, it means that the sparrows in the population are aware of the danger and need to approach other sparrows to minimize their risk of predation.

2.3 SSA–ESN model construction and algorithm steps

Since the regularization coefficients and learning rates of ESN are generally set empirically or randomly, the network structure has a certain degree of randomness. In this paper, the SSA algorithm is applied to the optimization of the regularization coefficient and learning rate to improve the prediction accuracy of the ESN model. Figure 2 demonstrates the network structure of SSA–ESN.

The steps of the SSA–ESN algorithm are as follows,

Step 1 input samples, normalize the data to the [0,1] interval;

Step 2 divide the training set and test set according to the ratio;

Step 3 choose the reservoir pool size, reservoir pool sparsity. Randomly generate W_{in} and W, and use ESN to train W_{out}. Set the value range of regularization factor and learning rate;

Step 4 optimize the regularization factor and learning rate using SSA to find their optimal values;

Step 5 return the optimal regularization coefficients and learning rate to the echo State Network and utilize it for prediction.

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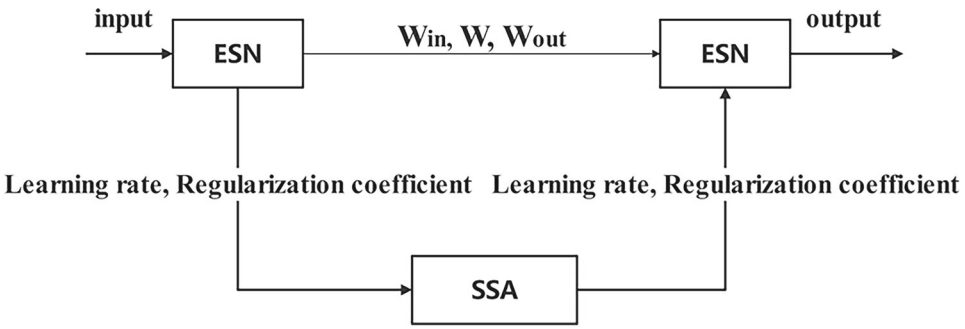


Fig. 2 Structural diagram of SSA–ESN

3 Model assumptions, data preprocessing and experimental configuration

3.1 Model assumptions

If the future has nothing to do with the past, and the patterns of the past will not repeat themselves in the future, then forecasting is all but impossible, and stock price forecasting is out of the question.

The model used in this paper is based on the following assumptions.

1. Stocks price show certain patterns, and past patterns will repeat themselves in the present and future.
2. Stocks price reflects the combined effects of national policies, stockholders' psychology, and corporate decisions on the stock market.

3.2 Data preprocessing

The data we obtain are sound, real and anomalous, so the data are not cleaned in this paper. However, in order to unify the basic measurement unit of data, speed up the convergence of the network and improve the generalization ability of the model, we carry out the normalization of the data. In fact, normalization of data is a common way of processing data. It maps the input data to a suitable range, avoids the gradient too large or too small, and helps to improve the stability of training.

3.3 Experimental configuration

Stocks price is affected by many factors, we uses the last day's opening price, highest price, lowest price, volume, turnover, closing price, the amplitude of the previous three days and its rise and fall of the price as indicators to predict present day's closing price (Tables 1, 2, 3).

Ten US stocks such as Intel (stock code: INTC) and Qualcomm (stock code: QCOM) and ten Chinese A-shares such as Vanke A (Stock Code: 000002) and Poly Development (stock code: 600048) are selected as research objects in this paper.

In order to verify the prediction accuracy of SSA–ESN model, this paper uses BP neural network, Elman neural network, LSTM and ESN to compare with it. Here, both ESN and SSA–ESN have a reservoir pool size of 100 and use the same weight matrix W_{in}, W.

Table 1 Ten selected representative US stocks

Stock name	Stock code	Stock name	Stock Code
Advanced Micro Devices	AMD	Intel	INTC
BNY Mellon	BK	Reiner Architecture	LEN
Horton House	DHI	LGI Home Inc	LGIH
Bank of America	BAC	NVIDIA Corporation	NVDA
Qualcomm	QCOM	Wells Fargo& Company	WFC

Table 2 Ten selected representative Chinese stocks

Stock name	Stock Code	Stock name	Stock Code
China Merchants Bank	600036	Vanke	000002
Poly Development Holding Group	600048	PingAn Bank	000001
Hangzhou Silan Microelectronics	600460	Allwinner Technology	300458
Bank of China	601988	Overseas Chinese Town	000069
Giga Device Semiconductor Inc	603986	Tongfu Microelectronics	002156

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Table 3 Model parameters of SSA–ESN and their values

Model	Parameters and their values	
	Parameter	Value
Echo state network	Reservoir pool size	100
	Learning rate	[0.001,2.0]
	Regularization coefficient	[0.0001,0.1]
	Example Initialize the reserve pool	30
Sparrow search algorithm	Sparrow number	10
	Maximum iterations	250
	Number of discoverers	2
	Aware of the number of dangerous sparrows	8
	Optimize the number of parameters	2
	Prewarning value	0.8

4 Results of using the SSA–ESN model to predict stock closing prices

For ten US stocks, such as Intel (stock code: INTC) and Qualcomm (stock code: QCOM), and ten Chinese A-shares, such as Vanke A (stock code: 000002) and Poly Development (stock code: 600,048), stock price prediction is firstly carried out by using the optimized echo state network of the sparrow search algorithm (Table 4).

We use stock data from January 15, 2021 to July 11, 2023. The first 400 days of data are used as the training set and the last 200 days are used as the test set. The test results are shown in Fig. 3 (Fig. 4).

From the above charts, it can be seen that whether it is the US stock market or the Chinese A-shares market, the stock closing price prediction method by ESN optimized by SSA works well, and it can provide effective help and guidance to the majority of stockholders for stock investment (Table 5).

5 Comparison of forecast results

In this section, the errors of BP neural network, Elman neural network, LSTM, ESN and SSA–ESN for multiple stocks prediction are compared to determine whether SSA–ESN is superior for solving the stock price prediction.

In this paper, four evaluation metrics are selected to evaluate the advantages and disadvantages among different models, including mean absolute error (MAE), mean square error (MSE), mean absolute percentage error (MAPE) and root mean square error (RMSE). They are defined as the following,

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|e_i|, \tag{9}$$

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(e_i)^2, \tag{10}$$

$$\mathrm{MAPE} = \frac{1}{N}\sum_{i=1}^{N}\frac{|e_i|}{|y_i|}, \tag{11}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(e_i)^2}. \tag{12}$$

5.1 Ten US stock predictions results

The BP neural network, Elman neural network, LSTM, echo state network and SSA–ESN models are used to predict the selected ten US stocks. The prediction results are displayed in Tables 6, 7, 8, 9 and 10 (Fig. 5).

It follows from the above graphs, for each US stock, SSA–ESN has the highest prediction accuracy among the five models.

5.2 Ten Chinese A-shares prediction results

The BP neural network, Elman neural network, LSTM, ESN and SSA–ESN model are used to predict the selected ten Chinese A-shares, respectively, and the prediction results are presented in Tables 11, 12, 13, 14 and 15 (Fig. 6).

It follows from the above graphs, for each Chinese A-share, SSA–ESN has the highest prediction accuracy among all the five models.

5.3 Robustness checks

In order to test the performance of the SSA–ESN model in different time frames and different market conditions, we select three typical Chinese stocks from the banking industry, real estate and semiconductor industry as research objects to observe the change of the prediction accuracy of the SSA–ESN model. The Chinese market is greatly affected by the novel coronavirus, so it will be a meaningful study to observe the prediction performance of the SSA–ESN model on December 9, 2022. We select two periods before and after China's critical point of containment of novel coronavirus, December 9, 2022, to conduct research. The ratios in the table below represent the ratio of the training set to the test set.

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Table 4 Test set errors for ten US stocks

Stoke code	RMSE	MAPE	Stoke code	RMSE	MAPE
AMD	2.883	0.027	INTC	0.768	0.021
BK	0.807	0.014	LEN	1.895	0.016
DHI	1.832	0.016	LGIH	2.899	0.022
BAC	0.619	0.015	NVDA	8.955	0.031
QCOM	3.026	0.020	WFC	0.849	0.015

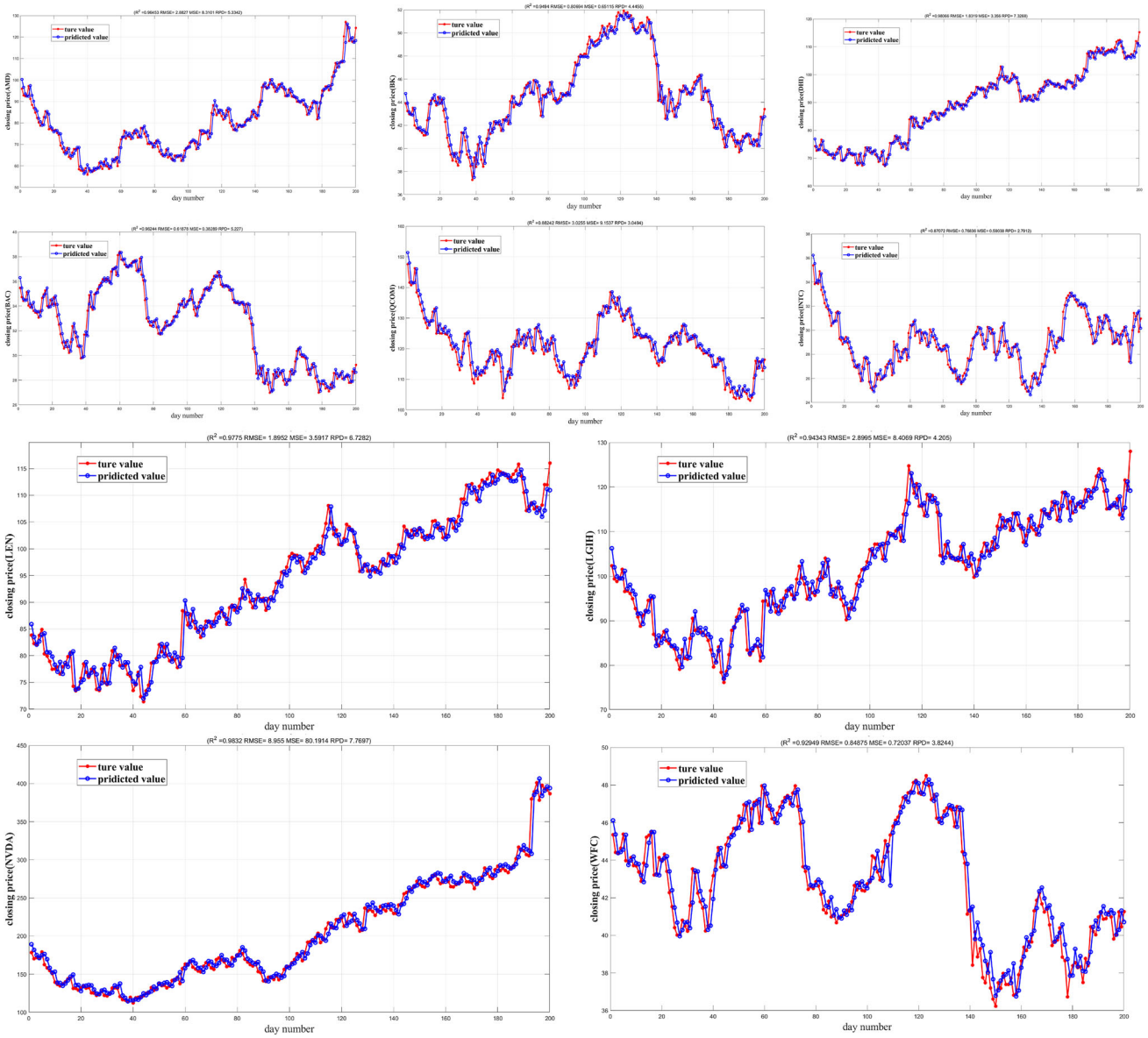


Fig. 3 Prediction of the test sets of ten US stocks

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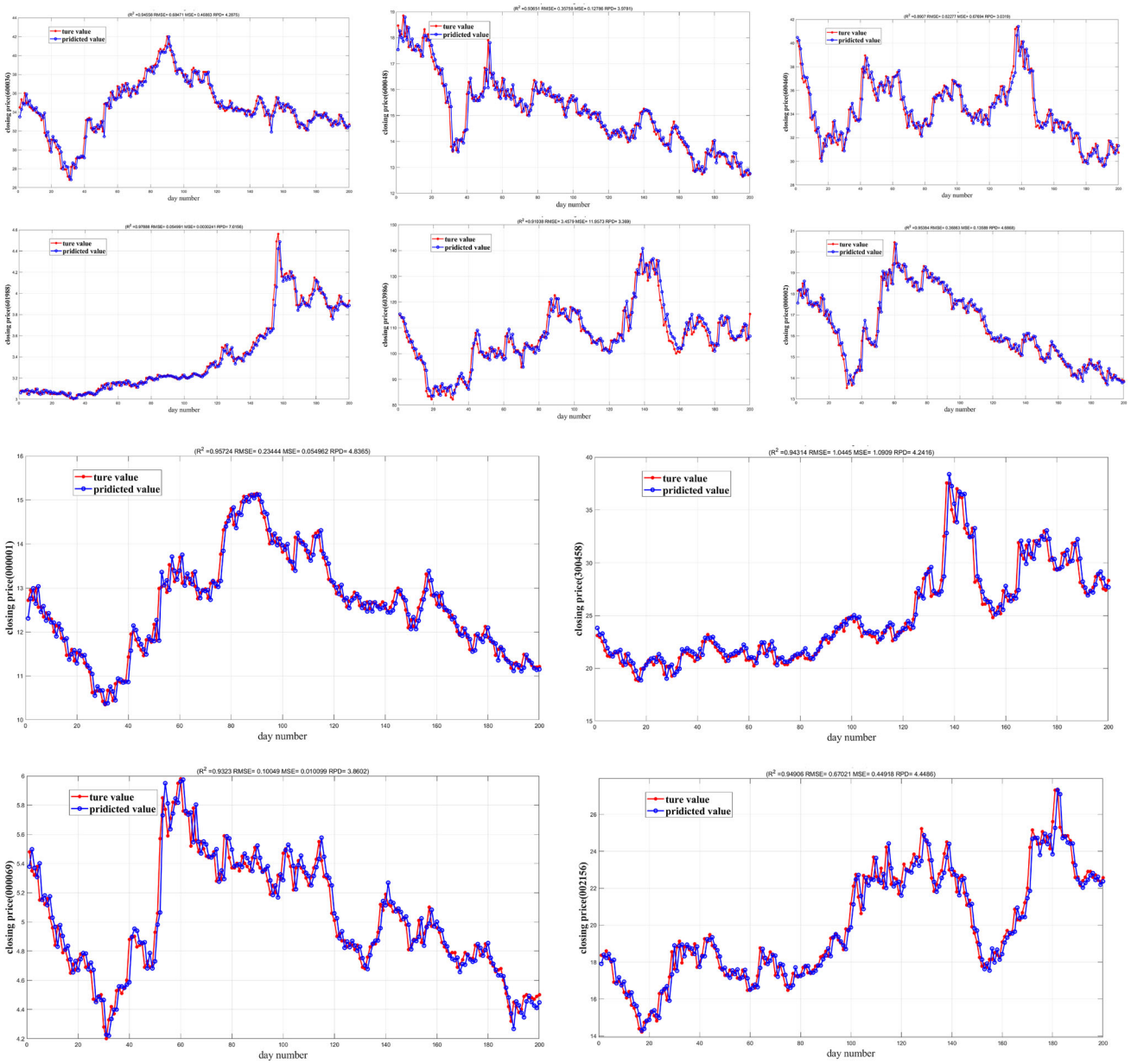


Fig. 4 Prediction of the test sets of ten A-shares in China

Table 5 Test set errors for ten A-shares in China

Stoke code	RMSE	MAPE	Stoke code	RMSE	MAPE
600036	0.685	0.015	000002	0.369	0.016
600048	0.358	0.016	000001	0.234	0.014
600460	0.823	0.018	300458	1.044	0.026
601988	0.055	0.009	000069	0.100	0.015
603986	3.458	0.024	002156	0.670	0.024

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Table 6 Prediction results of AMD and BK by BPNN, Elman, LSTM, ESN and SSA–ESN

	AMD					BK				
	BPNN	Elman	LSTM	ESN	SSA–ESN	BPNN	Elman	LSTM	ESN	SSA–ESN
MSE	17.48	24.63	78.33	9.196	8.310	0.936	1.323	1.935	0.659	0.651
MAE	3.229	3.861	6.661	2.277	2.128	0.713	0.912	1.031	0.631	0.627
RMSE	4.181	4.963	8.851	3.033	2.883	0.967	1.150	1.391	0.812	0.807
MAPE	0.038	0.054	0.071	0.029	0.027	0.016	0.021	0.022	0.014	0.014

Table 7 Prediction results of DHI and BAC by BPNN, Elman, LSTM, ESN and SSA–ESN

	DHI					BAC				
	BPNN	Elman	LSTM	ESN	SSA–ESN	BPNN	Elman	LSTM	ESN	SSA–ESN
MSE	4.468	5.226	11.57	3.872	3.356	1.612	37.404	2.993	0.417	0.383
MAE	1.637	1.727	2.625	1.532	1.399	0.873	5.601	1.374	0.489	0.466
RMSE	2.114	2.286	3.401	1.968	1.832	1.270	6.116	1.73	0.646	0.619
MAPE	0.018	0.019	0.030	0.017	0.016	0.025	0.181	3.943	0.015	0.015

Table 8 Prediction results of QCOM and INTC by BPNN, Elman, LSTM, ESN and SSA–ESN

	QCOM					INTC				
	BPNN	Elman	LSTM	ESN	SSA–ESN	BPNN	Elman	LSTM	ESN	SSA–ESN
MSE	46.68	57.108	99.58	12.68	9.154	6.945	7.420	61.04	0.701	0.590
MAE	5.809	6.143	8.221	2.943	2.386	2.289	2.461	7.574	0.668	0.597
RMSE	6.832	7.557	9.98	3.560	3.026	2.635	2.724	7.813	0.837	0.768
MAPE	0.047	0.053	0.065	0.025	0.020	0.077	0.088	0.257	0.023	0.021

Table 9 Prediction results of LEN and LGIH by BPNN, Elman, LSTM, ESN and SSA–ESN

	LEN					LGIH				
	BPNN	Elman	LSTM	ESN	SSA–ESN	BPNN	Elman	LSTM	ESN	SSA–ESN
MSE	7.935	5.096	8.127	4.186	3.592	14.525	25.845	26.96	8.906	8.407
MAE	2.183	1.752	2.235	1.571	1.428	3.067	3.841	4.126	2.284	2.231
RMSE	2.817	2.258	2.851	2.046	1.895	3.811	5.084	5.192	2.984	2.899
MAPE	0.023	0.019	0.024	0.017	0.016	0.029	0.040	0.038	0.023	0.022

Table 10 Prediction results of NVDA and WFC by BPNN, Elman, LSTM, ESN and SSA–ESN

	NVDA					WFC				
	BPNN	Elman	LSTM	ESN	SSA–ESN	BPNN	Elman	LSTM	ESN	SSA–ESN
MSE	121.9	268.0	859.72	146.6	80.19	1.597	1.127	1.868	0.758	0.720
MAE	7.538	11.48	22.33	8.879	6.085	0.883	0.795	1.083	0.646	0.628
RMSE	11.04	16.37	29.32	12.11	8.955	1.264	1.062	1.367	0.871	0.849
MAPE	0.034	0.058	0.082	0.044	0.031	0.020	0.019	2.455	0.015	0.015

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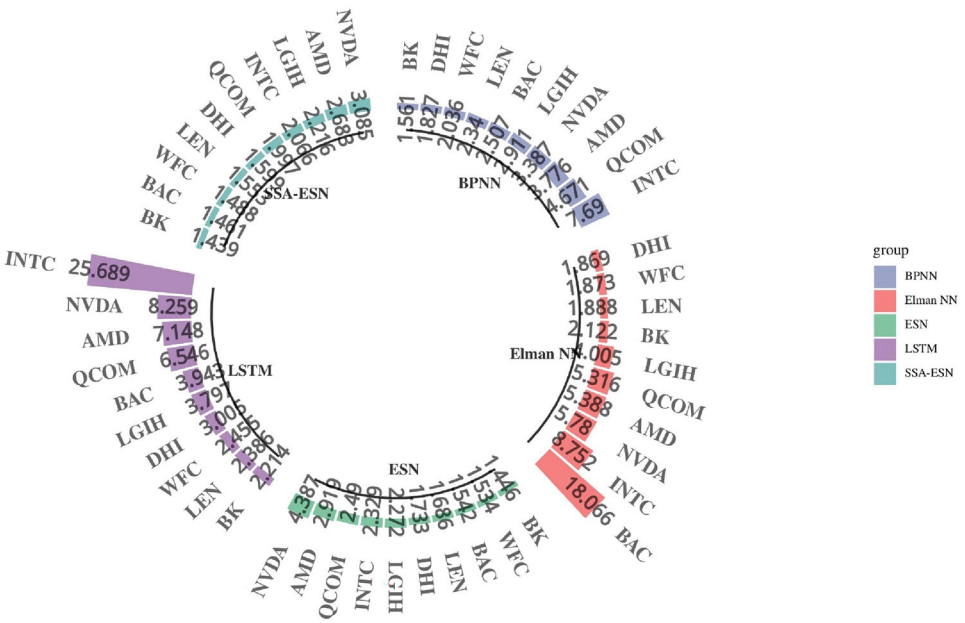


Fig. 5 MAPEs (%) of the test sets of ten US stocks predicted by each model

Table 11 Prediction results of 600036 and 600048 by BPNN, Elman, LSTM, ESN and SSA–ESN

	600036					600048				
	BPNN	Elman	LSTM	ESN	SSA–ESN	BPNN	Elman	LSTM	ESN	SSA–ESN
MSE	0.815	1.293	4.649	0.484	0.469	0.189	0.145	0.203	0.190	0.128
MAE	0.679	0.809	1.5323	0.511	0.509	0.289	0.265	0.316	0.347	0.248
RMSE	0.903	1.137	2.156	0.696	0.685	0.435	0.381	0.450	0.436	0.358
MAPE	0.019	0.025	0.041	0.015	0.015	0.020	0.017	0.021	0.023	0.016

Table 12 Prediction results of 600460 and 601988 by BPNN, Elman, LSTM, ESN and SSA–ESN

	600460					601988				
	BPNN	Elman	LSTM	ESN	SSA–ESN	BPNN	Elman	LSTM	ESN	SSA–ESN
MSE	1.893	1.151	1.296	1.012	0.677	0.105	0.138	0.070	0.003	0.003
MAE	0.938	0.817	0.831	0.812	0.619	0.181	0.210	0.152	0.031	0.031
RMSE	1.376	1.073	1.138	1.006	0.823	0.324	0.371	0.265	0.056	0.055
MAPE	0.028	0.024	0.025	0.024	0.018	0.061	0.054	0.051	0.009	0.009

Table 13 Prediction results of 603986 and 000002 by BPNN, Elman, LSTM, ESN and SSA–ESN

	603986					000002				
	BPNN	Elman	LSTM	ESN	SSA–ESN	BPNN	Elman	LSTM	ESN	SSA–ESN
MSE	73.37	58.68	255.57	15.19	11.957	0.233	7.399	1.596	0.170	0.136
MAE	6.667	6.065	13.748	3.061	2.528	0.333	2.307	1.018	0.328	0.262
RMSE	8.566	7.660	15.987	3.898	3.458	0.483	2.720	1.263	0.413	0.369
MAPE	0.059	0.059	0.121	0.029	0.024	0.020	0.151	0.058	0.020	0.016

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Table 14 Prediction results of 000001 and 300458 by BPNN, Elman, LSTM, ESN and SSA–ESN

	000001					300458				
	BPNN	Elman	LSTM	ESN	SSA–ESN	BPNN	Elman	LSTM	ESN	SSA–ESN
MSE	0.308	0.265	0.727	0.070	0.055	4.049	1.752	5.699	1.669	1.091
MAE	0.477	0.365	0.637	0.197	0.178	1.366	0.986	1.841	0.980	0.684
RMSE	0.555	0.514	0.853	0.264	0.234	2.012	1.324	2.387	1.292	1.044
MAPE	0.038	0.030	0.047	0.015	0.014	0.057	0.039	0.074	0.038	0.026

Table 15 Prediction results of 000069 and 002156 by BPNN, Elman, LSTM, ESN and SSA–ESN

	000069					002156				
	BPNN	Elman	LSTM	ESN	SSA–ESN	BPNN	Elman	LSTM	ESN	SSA–ESN
MSE	0.052	0.105	0.344	0.025	0.010	0.843	0.565	0.952	0.550	0.449
MAE	0.183	0.284	0.507	0.135	0.074	0.685	0.571	0.748	0.554	0.495
RMSE	0.228	0.324	0.587	0.158	0.100	0.918	0.752	0.976	0.741	0.670
MAPE	0.035	0.059	0.096	0.027	0.015	0.035	0.028	0.038	0.027	0.024

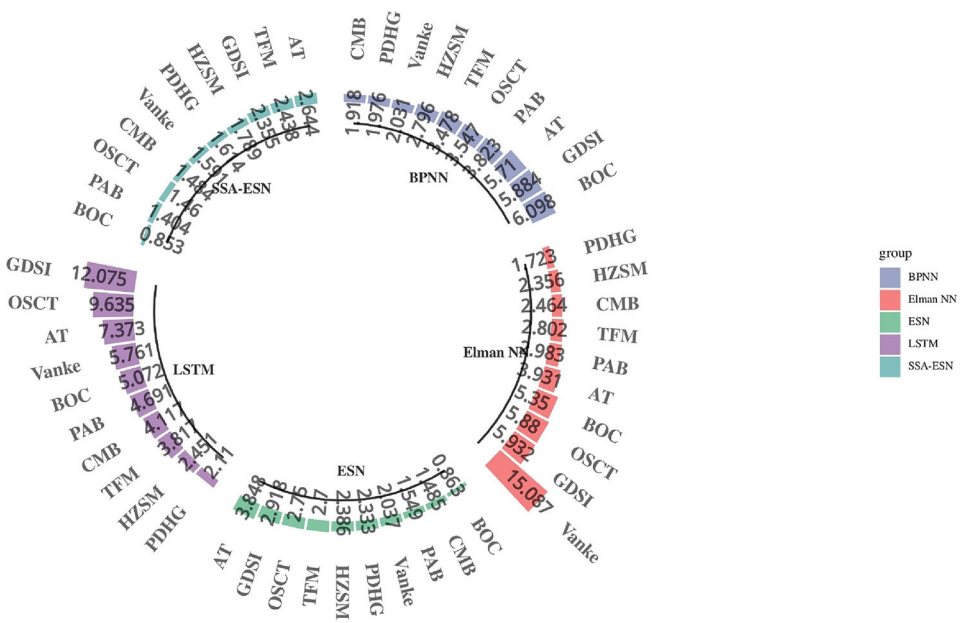


Fig. 6 MAPEs (%) of the test set of ten Chinese A-shares predicted by each model

Table 16 Forecast results of 000001 under different time frames and market conditions

	January 15, 2021–December 8, 2022				December 9, 2022-July 11, 2023			
	Train set		Test set		Train set		Test set	
	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE
1:1	0.645	0.023	0.319	0.017	0.264	0.016	0.206	0.014
2:1	0.591	0.022	0.291	0.015	0.271	0.016	0.173	0.011
6:4	0.708	0.028	0.291	0.015	0.276	0.016	0.164	0.010
8:2	0.765	0.034	0.213	0.013	0.273	0.016	0.142	0.009

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Table 17 Forecast results of 000069 under different time frames and market conditions

	January 15, 2021–December 8, 2022				December 9, 2022–July 11, 2023			
	Train set		Test set		Train set		Test set	
	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE
1:1	0.200	0.020	0.176	0.021	0.118	0.018	0.085	0.014
2:1	0.247	0.025	0.166	0.020	0.119	0.017	0.062	0.010
6:4	0.236	0.024	0.168	0.020	0.131	0.019	0.069	0.011
8:2	0.290	0.032	0.110	0.015	0.127	0.019	0.055	0.009

Table 18 Forecast results of 600,460 under different time frames and market conditions

	January 15, 2021–December 8, 2022				December 9, 2022–July 11, 2023			
	Train set		Test set		Train set		Test set	
	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE
1:1	1.990	0.034	1.345	0.022	0.945	0.021	0.075	0.016
2:1	1.873	0.031	1.344	0.023	0.930	0.020	0.751	0.017
6:4	1.880	0.032	1.362	0.023	0.906	0.020	0.779	0.017
8:2	1.772	0.029	1.154	0.020	1.056	0.024	0.679	0.017

Tables 16, 17 and 18 describe that the MAPE of SSA–ESN model for different industries is about 0.02 or less, indicating that its prediction performance is robust under different time frames and market conditions. In addition, we can note that the prediction performance of the test set is better, because the echo state network can capture long-term dependencies in the data by introducing memory units, which helps to improve the generalization ability of the model. It means that the model can make better predictions on data that has never been seen before.

6 Conclusion

The prediction of stocks prices is a major concern for a wide range of financial practitioners. The use of models with higher prediction accuracy facilitates more accurate financial decisions. A large amount of the literatures indicate that neural networks have higher prediction accuracy compared to traditional stocks price prediction methods. However, traditional neural network models have two major drawbacks for accurately predicting stocks price: one is that traditional neural networks are very sensitive to noise, and the other is that the initial parameters of traditional neural networks are often set empirically, and the randomness tends to have a great impact on the prediction accuracy. To address the first drawback of traditional neural networks, in this paper we use ESN to adapt to the high noise and severe nonlinearity of the stocks price market. For the second disadvantage of traditional neural networks, in this paper we use computational intelligence, combine the sparrow search algorithm with ESN, and optimizes its initial parameters, in order to avoid the decrease in prediction accuracy of traditional neural networks due to the randomness of the initial parameters.

In this paper, we investigate the prediction ability of the combined sparrow search algorithm and echo state network model in the stock market. BP neural network, Elman neural network, LSTM, ESN and SSA–ESN are compared. The prediction results of 20 active stocks in China and the USA prove that ESN has more accurate prediction results in stocks price prediction, and the use of the sparrow search algorithm can effectively improve the prediction performance of ESN.

The innovation of this paper is that the sparrow search algorithm and echo state network are successfully combined to deal with the stock price prediction with nonlinearity and high noise, and the prediction accuracy of echo state network is improved. At the same time, SSA–ESN model also inherits the advantages of echo state network, such as strong generalization ability, good stability, short running time, simple and effective model.

Financial practitioners can formulate investment strategies according to the forecast results of SSA–ESN, and can also use SSA–ESN model to evaluate the optimal strategy of the given investment strategy. In addition, the SSA–ESN model can also be combined with existing financial analysis tools. Here, are a few ways of combination:

1. Feature engineering: consider the feature variables used by existing financial analysis tools and combine them

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with the feature variables used by the SSA–ESN. For example, existing financial analysis tools may use technical indicators (such as moving averages, relative strength indicators, etc.) as feature variables, which can be used as additional feature variables in the SSA–ESN.

2. Model integration: the SSA–ESN can be integrated with other predictive models in existing financial analysis tools. The prediction results of different models can be combined through voting, weighted average, etc., to improve the overall prediction accuracy.
3. Risk management: use the risk management module provided by existing financial analysis tools to measure and manage the risk of forecast outcomes. By analyzing the error, confidence interval and other risk indicators, the prediction results can be better evaluated and the corresponding risk management strategies can be adopted.

However, there are some limitations to this study. It will be further discussion to optimize more parameters and explore its internal mechanism. The combination of advanced algorithms and data processing methods, such as modal decomposition, will also become a major trend in the field of financial forecasting.

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Author contributions ZT was responsible for data collection, data analysis and the writing of the full text. LP provided the direction and method of the research, and provided some guiding suggestions for the paper.

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Availability of data and materials The data that support the findings of this study are available on request from the corresponding author upon reasonable request.

Declarations

Conflict of interest Authors have no conflict of interest to declare.

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